DES (Data Encryption Standard)

Aim

To implement DES, a 64-bit block cipher built on a 16-round Feistel network with a 56-bit effective key. A single block is encrypted with an initial permutation, 16 rounds of expansion / key mixing / S-box substitution / permutation, and a final permutation; decryption runs the same rounds with the subkeys reversed. Arbitrary text is handled in ECB mode with PKCS#5 padding, and the ciphertext is shown as hexadecimal.

Algorithm

```
Function EncryptBlock(block, key):
    subkeys[0..15] ← key schedule of key            // PC1, rotate halves, PC2
    block ← IP(block)                               // initial permutation
    (L, R) ← left and right 32 bits of block
    for round ← 0 to 15 do
        prevR ← R
        R ← L XOR Feistel(R, subkeys[round])
        L ← prevR
    preoutput ← (R, L)                              // note the final swap
    return FP(preoutput)                            // final permutation
Function Feistel(R, subkey):
    x ← E(R) XOR subkey                             // expand 32 to 48 bits, mix key
    y ← S-box substitution of x                     // 8 boxes, 6 bits in, 4 bits out
    return P(y)                                     // permutation
```

```
Function DecryptBlock(block, key):
    subkeys[0..15] ← key schedule of key            // same schedule as encryption
    block ← IP(block)
    (L, R) ← left and right 32 bits of block
    for round ← 0 to 15 do
        prevR ← R
        R ← L XOR Feistel(R, subkeys[15 - round])   // reversed:  K16, K15, ..., K1
        L ← prevR
    preoutput ← (R, L)
    return FP(preoutput)
```

For arbitrary text the block cipher is wrapped: pad the plaintext to a multiple of 8 bytes with PKCS#5, encrypt each 8-byte block independently (ECB), and print the bytes as hex. Decryption reverses this and strips the padding.

Output

Encrypt (single block) — the standard DES test vector

Field	Value (hex)
plaintext	0123456789ABCDEF
key	133457799BBCDFF1
ciphertext	85E813540F0AB405

The 16 rounds run in order and the block matches the published FIPS result.

Decrypt (single block) — same key, reversed subkeys

Field	Value (hex)
ciphertext	`85E813540F0AB405`
key	`133457799BBCDFF1`
plaintext	`0123456789ABCDEF`

Encrypt (text mode) — plaintext `HELLO`, key `secret_k`

`HELLO` is 5 bytes, so PKCS#5 pads it to 8 bytes (append three `0x03` bytes), making one block. That block is DES-encrypted and printed as hex.

Step	Value
plaintext bytes	`48 45 4C 4C 4F`
after padding	`48 45 4C 4C 4F 03 03 03`
ciphertext hex	`CE25474AD7A640DA`

Result: `HELLO` -> `CE25474AD7A640DA`

Decrypt (text mode) — ciphertext `CE25474AD7A640DA`, key `secret_k`

The hex is decoded to one 8-byte block, DES-decrypted, and the PKCS#5 padding (the trailing `0x03` bytes) is removed.

Step	Value
decrypted bytes	`48 45 4C 4C 4F 03 03 03`
after unpadding	`48 45 4C 4C 4F`
plaintext	`HELLO`

Result: `CE25474AD7A640DA` -> `HELLO`